

Slide 1 – Title Slide

“Hello everyone. My name is Gour Krishna Dey, Roll number MT24035, and this is my Assignment 4 presentation on benchmarking machine learning models against Large Language Models using In-Context Learning.

In this project, I evaluated four different learning approaches across three datasets – the Iris dataset, the Wine dataset, and the Adult Income dataset – and compared their performance, strengths, and limitations.”

Slide 2 – Dataset & Task Overview

“In this assignment, the core objective was to test how traditional machine learning models compare against an LLM when all of them are given the same prediction task.

The datasets vary in complexity:

- Iris is a simple, clean, well-separated dataset.
- Wine is medium complexity with 13 numerical attributes.
- Adult Income is a high-complexity real-world dataset with mixed categorical and numerical features.

The models evaluated are Gaussian Naive Bayes, Decision Tree, Multi-Layer Perceptron, and an LLM using few-shot or in-context learning.

The main idea is to observe how each learning paradigm performs as the dataset becomes more complex.”

Slide 3 – Model Architectures

“Before going into dataset-wise results, here is a quick overview of the model architectures.

Naive Bayes is a probabilistic model assuming feature-wise independence. All numerical features are modelled using a Gaussian distribution. It trains extremely fast and works well when the independence assumption roughly holds.

Decision Tree builds hierarchical binary splits using the Gini impurity criterion. It automatically performs feature selection and gives a very interpretable structure. I used depth-based pre-pruning to avoid overfitting.

MLP, or Multi-Layer Perceptron, is a feed-forward fully connected neural network. It learns non-linear boundaries, uses ReLU activation, and is regularized with L2 penalty. It can model complex interactions but takes more time to train.

And finally, the **LLM with In-Context Learning** doesn't actually train on the dataset. Instead, it predicts based only on examples provided in the prompt. This works like pattern recognition rather than learning statistics."

Slide 4 – Iris Dataset

"Starting with the Iris dataset.

This dataset has 150 samples with 4 numeric features across 3 species. The classes are very well separated, especially by petal length and petal width. Because the patterns are simple and linear, almost all models perform extremely well.

Model-wise:

- Naive Bayes works nicely here because Gaussian assumptions roughly hold.
- The Decision Tree with depth 5 captures clean splits on petal dimensions.
- The MLP with 100 neurons also learns the decision boundaries easily.
- And the LLM with just 3–5 examples in the prompt perfectly matches the patterns.

So in this dataset, everything performs at its best."

Slide 5 – Iris Results

"These are the results.

LLM actually performs perfectly here with 100% in all metrics.

Decision Tree is very close at 98%.

MLP and Naive Bayes also stay above 96%.

The key insight: Iris is a simple dataset, and because the boundaries are clean, even a pattern-based model like an LLM can generalize extremely well.”

Slide 6 – Wine Dataset

“Moving to the Wine dataset.

This dataset is more complex than Iris because it has 13 chemical attributes and the class differences are subtle. Some features, like flavanoids, phenols, and proline, are highly correlated, and models have to capture these relationships.

Architecturally:

- Naive Bayes still works because the chemical features fit Gaussian-like distributions.
- Decision Tree with depth 3 works well but may slightly overfit specific patterns.
- The MLP uses a smaller architecture of 50 hidden units and converges quickly.

However, LLM struggles. It finds it difficult to interpret high-dimensional numeric relationships purely from examples. Its predictions drop significantly.”

Slide 7 – Wine Results

“In the results:

Naive Bayes performs best at around 97%.

Decision Tree achieves perfect precision and recall but slightly lower accuracy due to minor overfitting.

MLP stays in the low 90s.

But LLM drops sharply to 70%.

This validates that LLMs are extremely sensitive to numeric precision and don’t naturally capture statistical structure across many correlated numeric features.”

Slide 8 – Adult Income Dataset

“Next is the Adult Income dataset, the most complex one.

It has over 32,000 samples, mixed categorical and numeric features, and strong feature interactions. Predicting whether someone earns more than 50k requires modeling the relationships between education, occupation, capital-gain, marital status, and more.

In terms of architectures:

- Naive Bayes struggles because its independence assumption breaks badly for correlated demographic features.
- Decision Tree performs the strongest because it can capture hierarchical splitting patterns between categorical values and numeric thresholds.
MLP does well but requires much longer training time.
- LLM fails to model categorical patterns reliably, and it tends to under-predict the high-income class."

Slide 9 – Adult Results

"In the Adult dataset:

Decision Tree achieves the best performance with accuracy around 86%.

MLP also performs competitively at around 85%.

LLM reaches similar accuracy but shows very poor recall – only 58% – meaning it misses many high-income individuals.

Naive Bayes stays around 82%.

So, although LLM reaches high accuracy, it is biased toward predicting the majority class and fails to capture the complexity of tabular categorical interactions."

Slide 10 – Comparative Analysis

"Putting all datasets together:

Decision Tree is the most consistent performer – 98% on Iris, 95% on Wine, and 86% on Adult.

MLP is also consistent but slower to train.

Naive Bayes is efficient but performs poorly on datasets with correlated features.

LLM shows extreme variation — perfect on Iris, poor on Wine, and inconsistent on Adult.

One clear observation is that as dataset complexity increases, LLM performance declines, while ML models continue to adapt because they learn statistical structure from the data.”

Slide 11 — Code Walkthrough (Quick)

“In the code slide, I will briefly walk through the core components — loading datasets, preprocessing, training the models, evaluating metrics, and generating plots.

I won’t go into too much depth here because the full code and detailed explanations are included in the report and the GitHub repository. Anyone reviewing the code will be able to follow the modular structure and the evaluation pipelines.”

Slide 12 — Final Conclusion

“To conclude:

Decision Tree is the overall best model due to its balance of accuracy, interpretability, speed, and generalization. It performs reliably across simple, medium, and complex datasets.

MLP is powerful but computationally heavier.

Naive Bayes is fast but only suitable when independence assumptions are reasonable.

LLMs show promise for simple pattern-based classification but are not yet reliable for real-world tabular data.

In summary, traditional machine learning models remain superior for structured datasets, while LLMs can support analysis but cannot replace learned statistical models in tabular classification tasks.

This wraps up my presentation. Thank you.”